Due Date

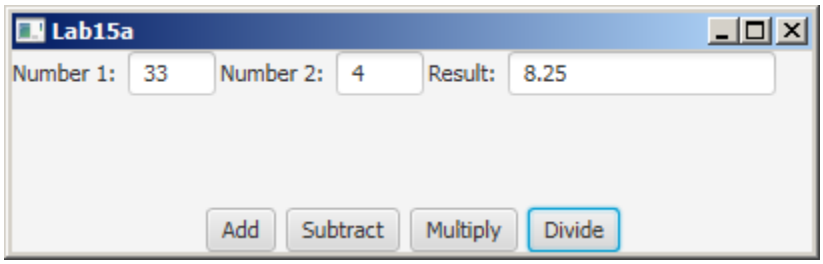
You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Wednesday, November 12, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Create a Simple Calculator

Write a JavaFX application which performs addition, subtraction, multiplication, and division, as shown in the image below:



- The user enters floating point numbers in the text fields labeled "**Number 1:**" and "**Number 2:**".
- If the user clicks on the **Add** button, then the program adds the two numbers, displaying the result in the "**Result:**" field.
- If the user clicks the **Subtract** button, then the value placed in **Result** is **Number1** – **Number2**.
- If the user clicks the **Multiply** button, then the value placed in **Result** is **Number1** * **Number2**.
- If the user clicks the **Divide** button, then the value placed in **Result** is **Number1** / **Number2**.

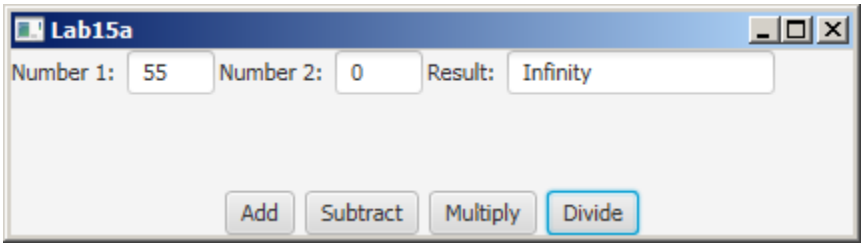
HINT:

- Create a **FlowPane** to hold the **Labels** and **Text** fields.
- Create a **HBox** to hold the **Buttons**.
- Create a **BorderPane** for the overall display.
- Add the **FlowPane** to the center region of the **BorderPane**.
- Add the **HBox** to the bottom region of the **BorderPane.**

Use the example programs provided in the **Sample Code** section of *Moodle* as a guide.

Special Case: Division by zero

Because we are using floating point variables (rather than integers) for this program, division by zero does not cause a runtime exception. Instead, the result is a special value called "**Infinity**":



If we were to write this program using integer variables, **then** dividing by zero would result in a runtime exception:

```
java.lang.ArithmeticException: / by zero
```

As always, submit the source code file to ***Moodle***, and demonstrate your solution during class.

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